Statistical Inference

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Inference

Definition 0.1 (Statistical inference)**. Statistical inference** is the process of analyzing data to learn about the shape and structure of the probability distribution that generated those data.

This definition is adapted from Wikipedia's [1].

Example 0.1 (Inference about cholesterol in the WCGS population)**.** The WCGS measured total cholesterol in 3,142 of its 3,154 participants. Using those measurements to learn about the distribution of cholesterol among men like the participants, for example its population mean, and quantifying how uncertain that knowledge is, is statistical inference.

Statistical inference typically consists of two steps:

1. fitting a statistical model to data;
2. summarizing our uncertainty about the parameters of the fitted model, based on the data (and, in Bayesian inference, on prior beliefs).

There are two predominant paradigms for statistical inference:

- frequentist inference, which treats parameters as fixed unknown constants;
- Bayesian inference, which treats parameters as random variables with prior distributions.

Hypothesis tests

Hypotheses

Definition 0.1 (Null hypothesis)**.** The **null hypothesis**, written H_0, is a specific claim about the parameters of a statistical model, which a hypothesis test assesses against the data.
Source: Wikipedia contributors [2].

Definition 0.2 (Alternative hypothesis)**.** The **alternative hypothesis**, written H_1 or H_A, is the claim that a hypothesis test weighs the data as evidence *for*, against the null hypothesis.

Definition 0.3 (Two-sided alternative)**.** For a null hypothesis that a parameter equals a value, such as $H_0 : \theta = \theta_0$, the **two-sided alternative** is that the parameter differs from that value in either direction: $H_1 : \theta \neq \theta_0$.
Source: Wikipedia contributors [3].

Example 0.1 (Cholesterol and CHD in the WCGS)**.** Let μ_1 and μ_0 be the population mean cholesterol of men who do and do not have a CHD event by 1969, in the population the WCGS sampled. A comparison of the two groups tests the null hypothesis that the population means are equal against the two-sided alternative that they differ:

$$H_0 : \mu_1 = \mu_0 \qquad H_1 : \mu_1 \neq \mu_0$$

p-values

Definition 0.4 (Test statistic)**.** A **test statistic** T is a function of the data whose distribution under the null hypothesis is known, exactly or approximately, and whose larger values are more extreme, meaning less consistent with H_0 and more consistent with H_1.
Source: Wikipedia contributors [4].

Definition 0.5 (p-value)**.** Given a test statistic T whose observed value is t, the **p-value** is the probability that T is at least as extreme as the value observed, computed with the data distributed as the null hypothesis specifies:

$$p \overset{\text{def}}{=} \Pr_{H_0}(T \geq t)$$

If the null hypothesis allows more than one distribution (a *composite* null hypothesis, such as $H_0 : \mu_1 = \mu_0$ with unknown variances), the p-value is the largest such probability over those distributions, or an approximation to it.
Source: Wikipedia contributors [5].

ℹ Remark (A p-value is not the probability that the null hypothesis is true)

Remark 0.1 (A p-value is not the probability that the null hypothesis is true). A p-value is computed assuming H_0 is true, so it is not the probability that H_0 is true given the data.

Definition 0.6 (Significance level)**.** A hypothesis test with **significance level** α rejects the null hypothesis when the p-value is at most α.
Source: Wikipedia contributors [6].

ℹ Remark (Choosing a significance level)

Remark 0.2 (Choosing a significance level). $\alpha = 0.05$ is a common convention, not a rule. When H_0 is true and the p-value is computed exactly, a test with significance level α rejects H_0 with probability at most α; for a test based on an approximate p-value, such as a large-sample test, this holds only approximately.

Definition 0.7 (Statistically significant)**.** A result is **statistically significant** at level α if it leads a test with significance level α to reject the null hypothesis.

Example 0.2 (Testing whether population mean cholesterol differs by CHD status)**.** Continuing Example 0.1, let $\bar{x}_1$ and $\bar{x}_0$ be the two groups' sample mean cholesterol, with sample variances s_1^2 and s_0^2 and sample sizes n_1 and n_0. By the central limit theorem, when H_0 holds and the samples are large, the statistic

$$Z \overset{\text{def}}{=} \frac{\bar{x}_1 - \bar{x}_0}{\sqrt{s_1^2/n_1 + s_0^2/n_0}}$$

has approximately a standard Gaussian distribution. Extreme values in either direction are evidence for the two-sided H_1, so the test statistic is $T = |Z|$, and the p-value is $\Pr(|Z| \geq |z|) = 2\Phi(-|z|)$, where Φ is the standard Gaussian CDF.

```r
chol_by_chd <- split(wcgs$chol, wcgs$chd69) |>
  lapply(function(x) x[!is.na(x)])
x1 <- chol_by_chd[["Yes"]]
x0 <- chol_by_chd[["No"]]
diff_means <- mean(x1) - mean(x0)
se_diff <- sqrt(var(x1) / length(x1) + var(x0) / length(x0))
z <- diff_means / se_diff
p_value <- 2 * pnorm(-abs(z))
c(difference = diff_means, se = se_diff, z = z, p_value = p_value)
#>  difference          se           z     p_value
#> 2.58087e+01 3.17988e+00 8.11626e+00 4.80780e-16
```

The p-value is far below 0.05, so at significance level 0.05 the test rejects H_0: the data provide statistically significant evidence that population mean cholesterol differs between the two groups.

Watch

5MinuteAI explains hypothesis testing in five minutes, following the University of Washington's CSE 312 [7].

Read

- Hogg et al. [8, Sec. 8.1], pp. 361–368 covers tests about one mean.
- Hogg et al. [8, Sec. 8.6], pp. 403–409 covers power of a statistical test.
- Casella and Berger [9, Sec. 8.3], pp. 382–401 covers methods of evaluating tests.
- Rice [10, Sec. 9.2], pp. 331–336 covers the Neyman–Pearson paradigm.

Reference distributions

The tests on this page compare a test statistic with one of three families of distributions, each built from independent standard Gaussian random variables. Here, "independent" means mutually independent.

Definition 0.1 (Chi-square distribution)**.** Let $Z_1, \ldots, Z_k$ be independent random variables, each with the standard Gaussian distribution. The distribution of $\sum_{j=1}^{k} Z_j^2$ is the **chi-square distribution with k degrees of freedom**, written χ^2_k.
Source: Wikipedia contributors [11].

Example 0.1 (The 0.95 quantile of χ^2_1)**.** By Definition 0.1, a χ^2_1 random variable is Z^2 for a standard Gaussian Z. So $\Pr(Z^2 \leq c) = \Pr(-\sqrt{c} \leq Z \leq \sqrt{c})$, and the 0.95 quantile of χ^2_1 is the square of the 0.975 quantile of the standard Gaussian distribution:

```
c(chisq = qchisq(0.95, df = 1), gaussian_squared = qnorm(0.975)^2)
#>            chisq gaussian_squared
#>          3.84146          3.84146
```

Definition 0.2 (t-distribution (Student's t-distribution))**.** Let Z have the standard Gaussian distribution, let V have the χ^2_k distribution (Definition 0.1), and let Z and V be independent. The distribution of

$$\frac{Z}{\sqrt{V/k}}$$

is the **t-distribution with k degrees of freedom** (or **Student's t-distribution**), written t_k [12].

Example 0.2 (t quantiles approach Gaussian quantiles)**.** The 0.975 quantile of t_k is larger than the standard Gaussian's 0.975 quantile, 1.96, and approaches it as k grows:

```
k <- c(4, 9, 29, 99, 2760)
tibble::tibble(k = k, t_quantile = qt(0.975, df = k))
#> # A tibble: 5 x 2
#>       k t_quantile
#>   <dbl>      <dbl>
#> 1     4       2.78
#> 2     9       2.26
#> 3    29       2.05
#> 4    99       1.98
#> 5  2760       1.96
```

So t-based intervals and tests differ noticeably from Gaussian-based ones only in small samples.

Definition 0.3 (F-distribution)**.** Let U have the $\chi^2_{k_1}$ distribution, let V have the $\chi^2_{k_2}$ distribution, and let U and V be independent. The distribution of

$$\frac{U/k_1}{V/k_2}$$

is the **F-distribution with k_1 and k_2 degrees of freedom**, written F_{k_1,k_2}.

Example 0.3 (A squared t random variable has an F-distribution)**.** Let $T = Z/\sqrt{V/k}$ as in Definition 0.2. Then

$$T^2 = \frac{Z^2/1}{V/k},$$

where Z^2 has the χ^2_1 distribution (Definition 0.1) and is independent of V. So T^2 has the $F_{1,k}$ distribution (Definition 0.3), and the 0.95 quantile of $F_{1,k}$ is the square of the 0.975 quantile of t_k:

```
c(f = qf(0.95, df1 = 1, df2 = 9), t_squared = qt(0.975, df = 9)^2)
#>         f t_squared
#>   5.11736   5.11736
```

Confidence intervals

Definition 0.1 (Confidence interval)**.** A $100(1-\alpha)\%$ **confidence interval** for a parameter θ is a random closed interval $[L(X), U(X)]$ whose endpoints $L(X) \leq U(X)$ are statistics computed from the data X, such that for every possible value of θ:

$$\Pr_\theta(\theta \in [L(X), U(X)]) \geq 1 - \alpha$$

The probability is over repeated samples from the data-generating process, with θ held fixed. The interval is random and θ is not.
Source: [9], Definitions 9.1.1, 9.1.4 and 9.1.5.

Definition 0.2 (Coverage probability)**.** The **coverage probability** of an interval $[L(X), U(X)]$ for a parameter θ is $\Pr_\theta(\theta \in [L(X), U(X)])$, computed over repeated samples with θ held fixed. A $100(1-\alpha)\%$ confidence interval has coverage probability at least $1-\alpha$ for every θ.
Source: [9], Definition 9.1.4.

Definition 0.3 (Confidence coefficient)**.** The **confidence coefficient** of an interval $[L(X), U(X)]$ for a parameter θ is the infimum over θ of its coverage probability, $\inf_\theta \Pr_\theta(\theta \in [L(X), U(X)])$.
Source: [9], Definition 9.1.5.

Definition 0.4 (Exact confidence interval)**.** A $100(1-\alpha)\%$ confidence interval $[L(X), U(X)]$ is **exact** when its coverage probability equals $1-\alpha$ for every possible value of θ.

Definition 0.5 (Conservative confidence interval)**.** A $100(1-\alpha)\%$ confidence interval $[L(X), U(X)]$ is **conservative** when its coverage probability is at least $1-\alpha$ for every possible value of θ and larger than $1-\alpha$ for some possible value of θ.

Definition 0.6 (Confidence set)**.** A $100(1-\alpha)\%$ **confidence set** for a parameter θ is a random subset $C(X)$ of the parameter space, computed from the data X, such that for every possible value of θ:

$$\Pr_{\theta}(\theta \in C(X)) \geq 1-\alpha$$

A confidence interval is a confidence set whose values are closed intervals.
Source: [9, Sec. 9.1].

Example 0.1 (A conservative interval for a binomial proportion)**.** Let $X \sim \text{Bin}(n, p)$ with $n = 20$, and let $[L(X), U(X)]$ be the Clopper-Pearson 95% interval returned by `binom.test()`. Its coverage probability at each p is the sum of the binomial probabilities of the values x of X for which $[L(x), U(x)]$ contains p:

```
n <- 20
cp <- t(sapply(0:n, function(x) binom.test(x, n)$conf.int))
coverage <- function(p) {
  sum(dbinom(0:n, n, p)[cp[, 1] <= p & p <= cp[, 2]])
}
p_grid <- seq(0.01, 0.99, by = 0.01)
cover <- sapply(p_grid, coverage)
range(cover)
#> [1] 0.958611 0.994366
```

The smallest coverage probability on this grid is 0.959, and the largest is 0.994. The grid is coarse and the coverage probability jumps at the interval endpoints, so the smallest grid value is only an upper bound for the confidence coefficient. The Clopper-Pearson interval is built to have coverage probability at least 0.95 at every p, and here it is above 0.95 at all 99 grid values of p, so it is conservative.

Remark (Why most intervals are conservative)

Remark 0.1 (Why most intervals are conservative). Definition 0.1 requires the coverage probability to be at least $1-\alpha$ at every θ, which is the same as requiring the confidence coefficient to be at least $1-\alpha$. The coverage probability can depend on θ, and the confidence coefficient is its greatest lower bound, which the coverage probability need not attain at any θ. An exact interval has the same coverage probability at every θ. Discrete data take only countably many values, so they usually allow only a conservative interval, as Example 0.1 shows. A confidence set drops the requirement that the region be an interval.

Definition 0.7 (Approximate confidence interval)**.** An **approximate** $100(1-\alpha)\%$ confidence interval is an interval whose coverage probability is approximately $1-\alpha$, for example only in large samples.

Example 0.2 (Confidence interval for population mean cholesterol in the WCGS)**.** By the central limit theorem, the sample mean $\bar{X}$ of a large sample has approximately a Gaussian distribution with mean μ and standard error $\sigma/\sqrt{n}$, which we estimate by $s/\sqrt{n}$. So $\bar{x} \pm z_{0.975}\, s/\sqrt{n}$, where $z_{0.975} \approx 1.96$ is the 0.975 quantile of the standard Gaussian distribution, is an approximate 95% confidence interval for μ:

```
chol <- wcgs$chol[!is.na(wcgs$chol)]
se_chol <- sd(chol) / sqrt(length(chol))
mean(chol) + c(lower = -1, upper = 1) * qnorm(0.975) * se_chol
#>   lower   upper
#> 224.854 227.891
```

Definition 0.8 (Margin of error (radius))**.** The **margin of error** (or **radius**) of a confidence interval is half the interval's width.
Source: Wikipedia contributors [13].

Example 0.3 (Margin of error for population mean cholesterol)**.** In Example 0.2, the margin of error is $z_{0.975} \times s/\sqrt{n} \approx 1.52$ mg/dL.

Watch

5MinuteAI explains confidence intervals in five minutes, following the University of Washington's CSE 312 [7].

Read

- Hogg et al. [8, Sec. 7.1], pp. 307–313 covers confidence intervals for means.
- Casella and Berger [9, Sec. 9.2], pp. 420–439 covers methods of finding interval estimators.
- Rice [10, Sec. 9.3], pp. 337–338 covers duality of confidence intervals and hypothesis tests.

Further reading on confidence intervals

For more on confidence intervals:

- Anatomy of a confidence interval (PDF);
- https://www.youtube.com/watch?v=vq1KrE7gU5M

Interpretation of negative findings

If a confidence interval includes the null value, or a hypothesis test fails to reject the null hypothesis, that result does not *necessarily* mean that the null hypothesis is true. (When the interval and the test are built from the same statistic, with coverage $1 - \alpha$ and significance level α, the two results coincide.) So we should not interpret such results as "the odds (or risks, hazards, or means) are not significantly different". Instead, we should write something like "the data do not provide statistically significant *evidence* that the odds (or risks, hazards, or means) differ". Statistical significance is a property of evidence, not of the estimands.

P-values do not distinguish between absence of evidence and evidence of absence. Confidence intervals do:

- a narrow confidence interval that includes the null value is evidence of absence (of any effect large enough to matter);
- a confidence interval that includes the null value but also includes substantially non-null values represents absence of evidence.

Conversely, even statistically significant evidence of a non-null value does not mean the value is large enough to matter; that depends on what the estimand is. For example, we might have statistically significant evidence that an exercise program prolongs human lifespans by 20 seconds, but an effect that small would be negligible in practical terms.

Figure 1 sketches these scenarios.

See also [14, Sec. 3.7].

```r
ci_scenarios <- tibble::tribble(
  ~scenario, ~lower, ~upper,
  "not enough data: we know almost nothing", 0.2, 5,
  "absence of evidence", 0.5, 2,
  "\"trending toward significance\" (still absence of evidence)", 0.95, 2.5,
  "evidence of absence", 0.97, 1.03,
  "evidence of a negligible effect", 1.02, 1.1,
  "uncertain whether negligible or substantial", 1.05, 1.8,
  "evidence of a substantial effect", 1.7, 1.8,
  "substantial, but imprecise", 1.5, 4
) |>
  dplyr::mutate(scenario = factor(scenario, levels = rev(scenario)))

ci_scenarios |>
  ggplot2::ggplot() +
  ggplot2::aes(y = scenario, xmin = lower, xmax = upper) +
  ggplot2::geom_errorbarh(height = 0.3) +
  ggplot2::geom_vline(xintercept = 1, linetype = "dashed") +
  ggplot2::scale_x_log10() +
  ggplot2::labs(x = "Ratio estimand (log scale); null value = 1", y = NULL)
```

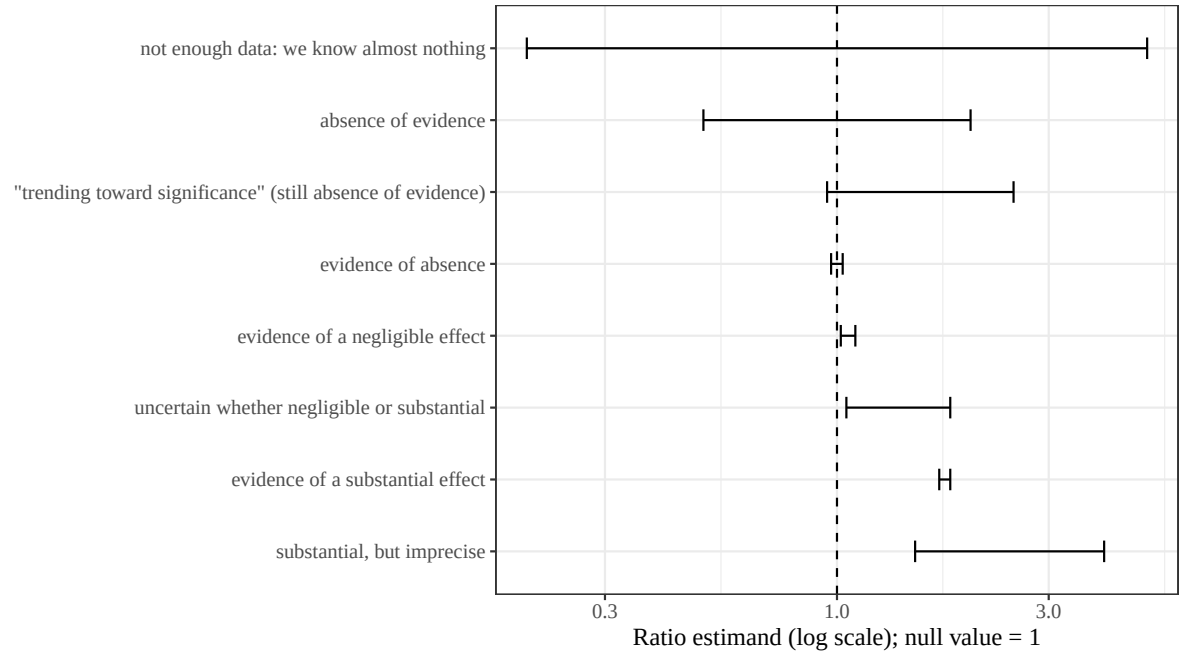


Figure 1: Interpretations of hypothetical 95% confidence intervals for a ratio (such as an odds ratio), relative to the null value 1 (dashed line). Adapted from a whiteboard sketch made in office hours.

References

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